

LLNL STEM Games Portal

Installation and User Guide

CSE 120 — Spring 2026 | UC Merced | Partner: Lawrence Livermore National Laboratory |

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Live: stemgames.online | GitHub: github.com/cse120-ucm/S26-CSE-335

Approved for public release by Dr. David Rakestraw, LLNL.

1. Product Overview

The LLNL STEM Games Portal (stemgames.online) is a free, browser-based platform hosting eight AI-powered STEM minigames for high school and undergraduate students. Features include AI tutors with voice and text interaction, a friends and leaderboard system, customizable profiles, and portal UI/game description translations in eleven languages (in-game content is in English). No download or installation is required for end users.

2. Platform and Dependencies

Dependency	Version	Purpose
Node.js	v18+ (v20 LTS recommended)	JavaScript runtime
npm	v9+ (bundled with Node)	Package manager
Git	v2+	Version control
Modern browser	Latest stable	Chrome, Firefox, Safari, or Edge

Built with Next.js (React + server-side API routes). Games use Vite; some use Phaser or KaPlay. External services: Supabase (DB/auth/storage), OpenAI (GPT-4o, Whisper), ElevenLabs (TTS), Upstash Redis (rate limiting), Cloudflare Workers (API proxy), Resend (email).

3. Installation

3.1 Clone and initialize

```
git clone https://github.com/cse120-ucm/S26-CSE-335.git
cd S26-CSE-335
git submodule update --init --recursive
```

3.2 Environment variables

Copy the template and fill in secret values:

```
cp .env.example .env.local
```

Variable	Service
NEXT_PUBLIC_SUPABASE_URL	Supabase
NEXT_PUBLIC_SUPABASE_ANON_KEY	Supabase
SUPABASE_SERVICE_ROLE_KEY	Supabase (server-side only)
OPENAI_API_KEY	OpenAI (GPT-4o + Whisper)
ELEVENLABS_API_KEY	ElevenLabs (TTS)
ELEVENLABS_VOICE_LAURIE	fUjY9K2nAlwlALowSiwc
ELEVENLABS_VOICE_LIVVY	B8gJV1lhpuegLxdpXFOE
UPSTASH_REDIS_REST_URL	Upstash Redis
UPSTASH_REDIS_REST_TOKEN	Upstash Redis
GAME_TOKEN_SECRET	32 character string used to hash the game tokens
RESEND_API_KEY	Resend (feedback emails)
FEEDBACK_FROM_EMAIL	Resend
FEEDBACK_RECIPIENT_EMAIL	Resend (fallback)
VITE_SUPABASE_ANON_KEY	Acceleration Sim Supabase Realtime Database
VITE_SUPABASE_URL	Acceleration Sim Supabase Realtime Database

Never prefix secrets with NEXT_PUBLIC_ — that exposes them to the browser.

3.3 Install, build games, and run

```
npm install
npm run setup-games
npm run dev
```

Portal starts at <http://localhost:3000>. The setup-games script builds all game submodules, copies outputs to `public/staticGames/`, and generates `src/data/games.ts`.

Production: `npm run build && npm start`. Deployed via Vercel with auto-deploys from main.

4. Adding a New Game

- Create a standalone web app that builds to `dist/` (Vite recommended, build command: `npm run build`).
- Add `data/game.json` (metadata) and `data/thumbnail.png` to the game repo.
- Set Vite base to `/staticGames/<game-id>/`.

- Add as submodule: `git submodule add <url> games/<game-id>` (folder must match game-id).
- Run `npm run setup-games` then `npm run dev`. Game appears on the home page.
- Portal UI/description translations generate automatically via Supabase seeding.

Games needing AI use portal proxy routes (POST `/api/ai/openai`, POST `/api/ai/openai/whisper`). No API keys in game code.

5. Using the Portal

- **Access:** Visit `stemgames.online` in any modern browser. Works on desktop and mobile. No download needed.
- **Account:** Sign up with email/password or Google OAuth to play games, save progress, and use social features.
- **Playing:** Click a game card on the home page. It loads in a sandboxed iframe. Fullscreen available.
- **AI Tutors:** Laurie-chan and Livvy-chan appear as visual novel overlays. Interact via hints, text chat, or hold-to-talk voice.
- **Social:** Add friends, view online status, compare scores. Per-game leaderboards (global + friends-only).
- **Profile:** Custom profile picture and banner with live preview.
- **Languages:** Portal UI and game descriptions in 11 languages via language picker. In-game content is in English.

6. Testing

Portal: `npm test` (Jest). Coverage: `npm run test:coverage`.

Games: `npx vitest run --coverage` from each game directory. Thresholds: 70–89%.

CI/CD: GitHub Actions runs tests on every push/PR. Deploy job builds submodules and deploys via Cloudflare Workers.